What Did Each Simulation Estimate?

AP Statistics · Topic 4.2 (CED 2.3) · B: 2026-10-20 · E: 2026-11-04 · TEACHER KEY

CED TETHER

- **Skill 3.A** — Determine relative frequencies, proportions, or probabilities using simulation or calculations.
- **EU UNC-2** — Simulation allows us to anticipate patterns in data.
- **LO UNC-2.A** — Estimate probabilities using simulation. [Skill 3.A]
- **EK UNC-2.A.1** — A random process generates results that are determined by chance.
- **EK UNC-2.A.2** — An outcome is the result of a trial of a random process.
- **EK UNC-2.A.3** — An event is a collection of outcomes.
- **EK UNC-2.A.4** — Simulation is a way to model random events, such that simulated outcomes closely match real-world outcomes. All possible outcomes are associated with a value to be determined by chance. Record the counts of simulated outcomes and the count total.
- **EK UNC-2.A.5** — The relative frequency of an outcome or event in simulated or empirical data can be used to estimate the probability of that outcome or event.
- **EK UNC-2.A.6** — The law of large numbers states that simulated (empirical) probabilities tend to get closer to the true probability as the number of trials increases.

Essential question: How must chance labels, one trial, and the success event align with the probability question?

DOK-3 skill: 4.B · **FRQ pattern:** `whose-simulation-design-is-valid-and-what-it-misestimates`

DOK rationale: Part (c) traces two different design mismatches to the events they actually estimate, weighs their competing effects and random variation, and repairs all parts of the simulation for the target question.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) **DOK:** 1 → 3 **Minutes:** 45

Teacher does

- Board slide up at the bell. Record Student 1, Student 2, and neither votes without resolving them.
- Begin the combined 4.1–4.2 video follow-along at minute 5.
- Return to the board slide at minute 33. Circulate and ask what probability each plan actually targets.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Choose a simulation result and name one design feature in a first take.
- Complete the follow-along about outcomes, events, chance assignments, trials, and relative frequency.
- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

Questions to ask

- What probability does one digit out of 10 represent?

- Which left-handed counts belong to the words at least 2?
- Can a reported estimate land near the target even when its design targets the wrong event?
- After your repair, can you point to the assignment, one trial, and the success event separately?

Adult role
Solo teacher. Circulate 5 pairs. Require students to translate each row into the probability it estimates before accepting which number looks closer.

[WARN] WATCH FOR

- Calling Student 2 valid because its 12 percent assignment is correct.
- Treating exactly 2 and at least 2 as interchangeable.
- Choosing the larger decimal without discussing design or simulation variability.

SCORING PART (c) — E / P / I

Expected elements

- **diagnoses-probability-mismatch** (required) — Explains that Student 1 assigns probability 0.10 rather than 0.12 to left-handed
- **diagnoses-event-mismatch** (required) — Explains that Student 2 estimates exactly 2 rather than at least 2 left-handed students
- **adjudicates-with-uncertainty** (required) — Identifies Student 2 as expected to be closer while explaining that wrong targets and simulation variability prevent either design from being trusted
- **repairs-entire-trial** (required) — Specifies five independent 00–99 pairs with 00–11 left-handed and counts at least two left-handed outcomes

E	Diagnoses both mismatches, gives a defensible Student 2 comparison with the appropriate uncertainty, and supplies a complete corrected assignment, five-selection trial, and at-least-2 success event.
P	Diagnoses both flaws but incompletely adjudicates the estimates or omits one element of the corrected trial, or gives a correct trial with only one flaw explained.
I	Calls either original design valid, confuses exactly with at least, or proposes a trial whose labels do not assign probability 0.12.

Common mistakes

- Counting digit 0 as 12 percent because 0 is one of the possible outcomes
- Treating exactly 2 and at least 2 as the same event
- Choosing Student 2 solely because two-digit numbers look more precise
- Fixing the label assignment but continuing to count exactly 2

[STAR] TODAY'S PROBLEM — annotated

Suppose each of five students can be treated as independently selected, and each has probability 0.12 of being left-handed. Two students use 1,000 simulation trials to estimate the probability that at least 2 of the 5 students are left-handed. Their plans and results are shown.

Results: Student 1 = 87/1,000; Student 2 = 94/1,000

Plan	L assignment	Trial; success
1	0 of 0–9	5 digits; 2+ L
2	00–11 of 00–99	5 pairs; exactly 2 L

FIRST TAKE (5 min)

Which simulation result would you trust more for the question asked—Student 1, Student 2, or neither? Name one reason.
Key: Not graded. Accept a tentative choice; the lesson should move students from decimal comparison to model-event alignment.

Rules callout "THREE MATCHES MAKE A VALID SIMULATION (keep on the page)" is printed on the student sheet.

DOK 1 **(a)** Write the event "at least 2 of 5 are left-handed" as a list of possible left-handed counts. State the probability that the random labels in a valid design must assign to left-handed.

Key: The event includes 2, 3, 4, or 5 left-handed students. A valid random-label assignment must designate 12 percent of equally likely labels as left-handed.

DOK 2 **(b)** Calculate each student's reported estimate. For each design, compare its left-handed assignment and its counted success event with the target question.

Key: Student 1 reports $87/1000 = 0.087$ and Student 2 reports $94/1000 = 0.094$. Student 1 assigns only $1/10 = 0.10$ of the digits to left-handed, although the success event is correctly at least 2. Student 2 assigns $12/100 = 0.12$ of the pairs to left-handed, but counts exactly 2 rather than 2, 3, 4, or 5.

DOK 3 **(c)** Identify the flaw in each design. Decide which reported estimate you expect to be closer to the target probability, and explain why two flawed, variable simulations cannot establish that choice reliably. Then write one correct trial description, including the random-number assignment and the event counted.

Key: Student 2's 0.094 is expected to be closer because its 0.12 assignment is exact and it omits only trials with 3, 4, or 5 left-handed students. Student 1 lowers every left-handed chance to 0.10. For verification, the desired probability is $1-(0.88)^5-5(0.12)(0.88)^4 = 0.1124508672$; Student 1's long-run target is $1-(0.90)^5-5(0.10)(0.90)^4 = 0.08146$, while Student 2's is $\binom{5}{2}(0.12)^2(0.88)^3 = 0.098131968$. Both designs systematically estimate the wrong quantity, and simulation-to-simulation variability means observed closeness cannot validate either one. A correct trial generates five independent pairs from 00–99, lets 00–11 represent left-handed and 12–99 represent right-handed, and counts a success when at least two of the five pairs represent left-handed. Repeat many trials and use the proportion of successful trials.

EXIT

One thing the video changed about my first take: ____________________